Module: reports and exports

English version of ../../modules/reports.md.

1. Purpose and boundaries

Key indicators, the distribution of interactions across stages, Excel/JSON exports and PDF guides. It reads interaction data and changes nothing in it. It does not: modify stages or cards; it is not the source of truth — that lives in the interactions module.

2. Use cases

- A report by stage and product for a period.
- Role-based filtering: an executive sees everything, a KAM only their own interactions.
- Excel export with correct Cyrillic; JSON export.
- Failure: an empty period → an empty report with an explicit zero, not an error.

3. Contract

`/api/reports/*`, `/api/reports/export/*`. Report columns are canonical and shared with the frontend: `config/report_columns.json` (synchronised by `scripts/sync_report_columns.py`).

4. Data

No tables of its own: aggregates are computed from interactions. The aggregate cache is Redis/Key-DB (`services/report_cache.py`); when the cache is unavailable the computation happens directly.

5. Permissions

The visibility scope of a report depends on the role (`tests/test_report_scope.py`): an executive sees everything, a KAM only their own. This is enforced by a test, not by convention.

6. Dependencies

Outbound: the database, the cache, the `openpyxl`/`python-pptx` export stack. Inbound: `api.reports`, the `ReportPage` page.

7. Configuration

Redis settings (`REDIS_URL`), export limits. With the cache disabled reports are slower but still correct.

8. Build and debugging

```
python -m pytest tests/test_report_columns.py tests/test_report_scope.py -q
python -m pytest tests/test_excel_export_cyrillic.py -q
```

9. Tests

`test_report_columns.py` (the canonical column set), `test_report_scope.py` (role → visibility), `test_excel_export_cyrillic.py` (Cyrillic in Excel), `test_json_export.py`, `test_pdf_layout.py`, `test_guide_pdf.py`.

10. Observability

Cache and export logs; no audit is written (a read operation). Errors: `report_scope_denied`, `export_failed`.

11. Limitations and readiness

- Reports are computed on request: large volumes need the cache.
- There is no ad-hoc report builder.
- Readiness: the visibility scope is covered by a test; Cyrillic does not break in Excel.